RF-Based Drone Identification Using Machine Learning

[23ET7PE3ML]

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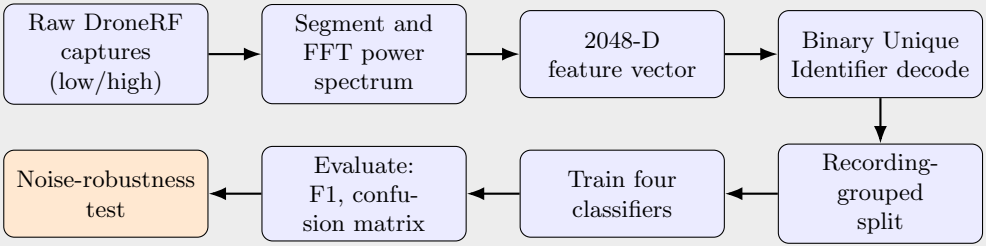
Presented by

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Introduction and Problem Definition

- Unmanned Aerial Vehicles (UAVs) transmit continuous **Radio Frequency (RF)** control signals to their operators, enabling **passive detection** without line of sight, unlike optical or radar sensing.
- These links operate in the **2.4 GHz ISM band** (Industrial, Scientific and Medical). This band is chosen because it is licence-free worldwide and balances range against antenna size; lower bands sacrifice data rate, higher bands sacrifice range and obstacle penetration.
- Detecting **presence** alone is insufficient; a practical system must also determine the **drone family** and its **operating condition** to assess threat level.
- The 2.4 GHz band is shared with **Wi-Fi and Bluetooth**, so recorded signals are noisy and classification is non-trivial.
- Segments extracted from a single recording are highly correlated; assigning them to both training and test sets causes **data leakage** and inflates measured accuracy.

Proposed Methodology



The **Fast Fourier Transform (FFT)** converts each time-domain segment into a frequency fingerprint; three labels are then decoded from a single filename code.

Dataset Used: DroneRF

Why this dataset: DroneRF is the standard public benchmark for RF-based drone detection, published with the founding study in this field, which makes results directly comparable with prior work.

Class (Family)	BUI codes	Modes	Role
Background (no drone)	00000	1	Negative class
Parrot Bebop	10000–10011	4	Drone family 1
Parrot AR	10100–10111	4	Drone family 2
DJI Phantom	11000	1	Drone family 3

Scale: approximately **41 GB** of raw recordings, comprising **227 recording pairs**, which yield **22,700** labelled feature vectors after segmentation.
BUI denotes a **Binary Unique Identifier**, the 5-digit code that labels every recording (expanded on the next slide).

Label Decoding from the Filename

Example files on disk:

```
10111H_3.csv
10111L_3.csv
# contents (RF samples)
-0.021, 0.034, -0.012,
0.045, 0.008, -0.030,
0.019, -0.007, 0.041, ...
```

BUI = Binary Unique Identifier (5 digits): the first digit marks drone presence, the next two the drone family, the last two the operating mode. The letter (H/L) marks the high or low RF sub-band.

All three labels decode from the code:

Target	Rule
Presence (2 classes)	0 if background, else 1
Family (4 classes)	background, Bebop, AR or Phantom
Mode (10 classes)	one index per BUI code

No manual labelling is required, since the ground truth is embedded in each filename.

Feature Extraction

Raw RF captures are long sequences of amplitude samples that appear as noise in the time domain. The drone's signature is visible only in the **frequency domain**, so the pipeline transforms each segment as follows:

1. **Segmentation:** each capture is divided into windows of $L = 100{,}000$ samples, long enough to hold a stable spectrum yet short enough to yield many training examples.
2. **Fast Fourier Transform:** a 2048-point FFT (a power of two, for computational efficiency) converts each window to the frequency domain; the upper 1024 bins are retained, as the magnitude spectrum is symmetric.

$$X_f = \left| \, \mathrm{FFT}_M(x - \bar{x}) \, \right|, \qquad f = \tfrac{M}{2} + 1, \ldots, M$$

3. **Band fusion:** the low (`L`) and high (`H`) sub-bands are concatenated and squared to a **power spectrum**, giving a single **2048-dimensional** vector per segment.

Result: a feature matrix of **22,700** $\times$ **2048** for model training.

Classification Models and Rationale

Model	Rationale for inclusion	Observed behaviour
Logistic Regression	Linear baseline; strong performance would imply a linearly separable task	Perfect on presence and family; weak on mode (0.375)
Random Forest	Bagging ensemble that captures non-linear structure	Stable; marginally lower on family (0.898)
Support Vector Machine (RBF kernel)	Models curved decision boundaries	Strong on simple tasks; limited on mode (5000-sample training cap)
XGBoost	Gradient-boosted trees; state of the art on tabular data	**Best overall**, and decisively best on the mode task (0.492)

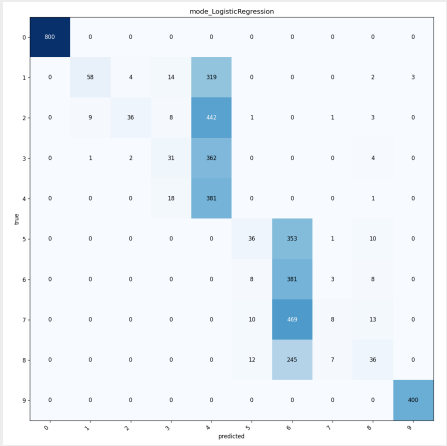
All four models perform near-identically on the easy tasks and separate only on the 10-class operating-mode task, where XGBoost leads. That task is therefore where the comparison is meaningful.

Classification Results

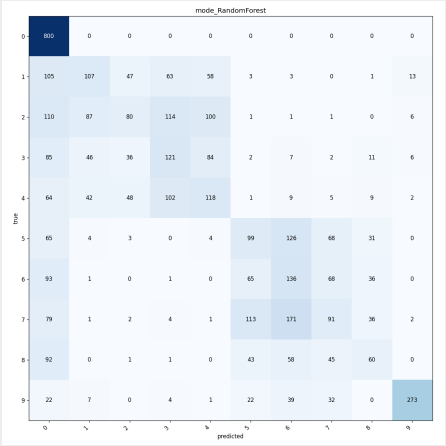
Model	Presence	Family	Mode (5-fold)
Logistic Regression	1.000	0.999	0.375
Random Forest	0.978	0.898	0.378
Support Vector Machine	1.000	0.977	0.160
XGBoost	**1.000**	**1.000**	**0.492**

- Scores are reported as **macro-averaged F1**, which is robust to the class imbalance in the dataset.
- **Presence** and **family** are essentially solved; the 10-class **operating-mode** task is hard, with XGBoost leading at 0.492 ($\pm$0.009 over five folds).

Confusion Matrices: Operating Mode

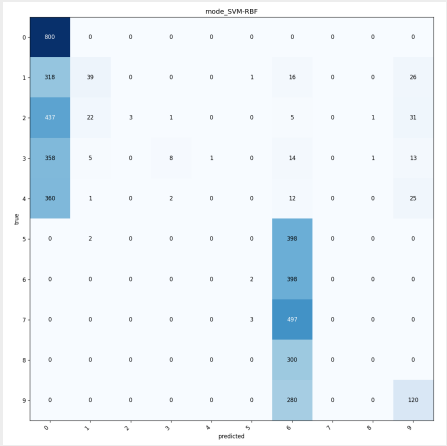


Logistic Regression

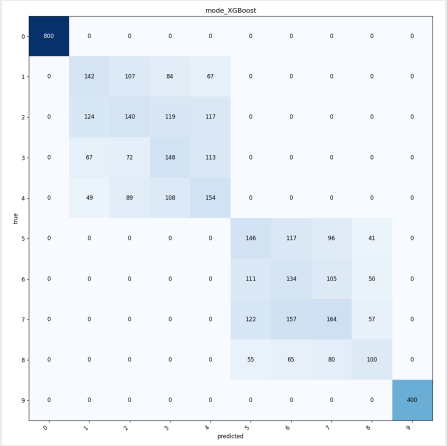


Random Forest

Confusion Matrices: Operating Mode

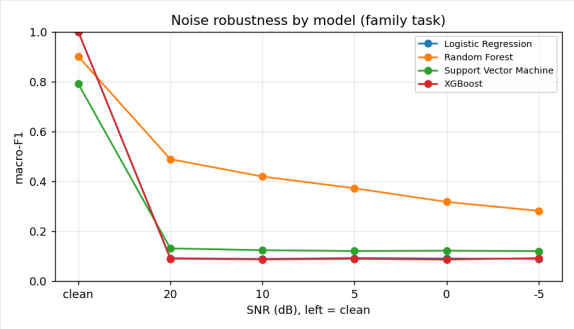


Support Vector Machine



XGBoost (strongest diagonal)

Robustness Under Noise and Interference



Method: each model is trained on clean data, then evaluated on the family task as **Gaussian noise** is added to the test set at decreasing **Signal-to-Noise Ratio (SNR)**.

Finding: Random Forest is the most robust, degrading gracefully. The models with the highest clean scores (XGBoost, Logistic Regression, Support Vector Machine) collapse to near-chance once any noise is added.

Implication: high clean accuracy does not imply robustness. A deployed system would favour Random Forest or require explicit noise handling.

Advantages and Limitations

Advantages

- **Passive detection** without line of sight or lighting constraints.
- **Interpretable, low-cost** models that train rapidly on standard hardware.
- **Leakage-aware** evaluation yields trustworthy metrics.
- A single pipeline answers all three classification questions.

Limitations

- **Operating-mode** classification remains difficult (best F1 0.492).
- Spectral features **degrade sharply** under heavy noise.
- Only **three drone families** are represented in the dataset.
- Dataset-based study; **no live hardware** or field deployment.

Applications and Conclusion

Applications

- **Passive surveillance** near airports and restricted zones.
- Early warning for unauthorised drones at public events.
- Low-cost monitoring on **software-defined radios**.
- Drone-type identification for **threat assessment**.

Conclusion

- Four classical models were compared across three tasks under a **leakage-aware** split.
- Presence and family are reliably classified; **XGBoost** leads the hard mode task at 0.492.
- A **noise-robustness** test quantified feature sensitivity to interference.
- Classical models remain **competitive with deep learning** on this dataset.

Literature Survey

Reference	**Approach**	**Relevance to the present work**
Al-Sa'd *et al.*, 2019	Deep Neural Networks	Established the DroneRF dataset and the three-task structure adopted here.
Allahham *et al.*, 2019	Dataset descriptor	Documents the file structure and the Binary Unique Identifier labelling scheme.
Al-Emadi *et al.*, 2020	XGBoost, three tasks	Closest prior study; corroborates XGBoost's strength and the difficulty of the mode task.
Ezuma *et al.*, 2019	k-Nearest Neighbours	Classical-ML baseline using statistical RF-fingerprint features.
Ezuma *et al.*, 2020	Machine learning under interference	Motivates the noise-robustness analysis in the shared 2.4 GHz band.

Literature Survey

Reference	**Approach**	**Relevance to the present work**
Medaiyese *et al.*, 2022	Wavelet-transform features	Principal alternative to the Fast Fourier Transform features used here.
Medaiyese *et al.*, 2022	Hierarchical learning	A hierarchical analogue of the presence, family and mode task structure.
Nemer *et al.*, 2021	Hierarchical ensemble	A close methodological sibling to the present classical-ML pipeline on DroneRF.
Basak *et al.*, 2022	Combined detection and classification	Studies robustness under multipath and varying Signal-to-Noise Ratio.
Swinney and Woods, 2021	Multi-channel deep network	Focuses on the operating-mode task, the hardest task in this project.

Literature Survey

Reference	Approach	Relevance to the present work
LUMEN, 2026	Power Spectral Density features	Validates the power-spectral feature choice on constrained hardware.
Complex-valued CNN, 2026	Low-SNR recognition	Recent baseline on a DroneRF-family dataset at low Signal-to-Noise Ratio.
Robustness study, 2024	RF detector robustness	Establishes Signal-to-Noise-Ratio degradation testing as standard practice.
Composite ensemble, 2024	Ensemble under jamming	Supports the Random Forest and XGBoost choice under interference.
RFUAV benchmark, 2025	Dataset with SNR tools	A larger benchmark and a direction for future work.

Research gap: prior studies are largely deep-learning-based or omit an explicit leakage check. This work compares four interpretable classical models across all three tasks under a recording-grouped split.

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